

- 5 A particle  $P$  is projected from a point  $O$  on a horizontal plane and moves freely under gravity. Its initial speed is  $u \text{ ms}^{-1}$  and its angle of projection is  $\sin^{-1}(\frac{4}{5})$  above the horizontal. At time 8 s after projection,  $P$  is at the point  $A$ . At time 32 s after projection,  $P$  is at the point  $B$ . The direction of motion of  $P$  at  $B$  is perpendicular to its direction of motion at  $A$ .

Find the value of  $u$ .

[7]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.